CS-471: Machine Learning

Week 4

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Multinomial Logistic Regression
- Classification Model Performance Evaluation
- Practical Aspects of Machine Learning

Recap of Previous Lecture

- What is the loss function in logistic regression?
- What is the gradient of above loss function?
- What is the update rule for logistic regression?
- How do we solve a linear classification problem using hinge loss based linear classifier that uses the sign function in hypothesis class?
- How do we solve a linear classification problem using logistic regression?
- How do we do multiclass classification using one vs rest approach? One vs one approach?

Multinomial Logistic Regression

- Each class has a separate weight vector $\boldsymbol{w_k}$.

- Can be combined to form a weight matrix $\boldsymbol{W}$:

$$\boldsymbol{W} = [\boldsymbol{w_1} \quad \boldsymbol{w_2} \quad \dots \quad \boldsymbol{w_K}]$$

- Where $\boldsymbol{K}$ is the total number of classes.

- Order of $\boldsymbol{W}$ is $(f + 1) \times K$ where f is the number of features.

Multinomial Logistic Regression

- Let the input matrix be $\boldsymbol{X}$:

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_B \end{bmatrix}$$

- B is the number of training examples in a batch.
- $\boldsymbol{x}_i$ is the feature vector associated with a particular input.
- Order of $\boldsymbol{X}$ is $B \times (f + 1)$

Multinomial Logistic Regression

- The score matrix can then be defined as $\boldsymbol{Z}$:

$$\boldsymbol{Z} = \boldsymbol{X}\,\boldsymbol{W}$$

- Order of $\boldsymbol{Z}$ is $B \times K$

- Each row of $\boldsymbol{Z}$ corresponds to score vector associated with a particular training example.

Multinomial Logistic Regression

- The probability matrix can then be defined as $\boldsymbol{P}$:

$$\boldsymbol{P} = \boldsymbol{softmax}(\boldsymbol{Z})$$

- Order of $\boldsymbol{P}$ is $B \times K$

- Softmax is applied row wise:

$$p_{ik} = \frac{e^{z_{ik}}}{\sum_{j=1}^{K} e^{z_{ij}}}$$

Multinomial Logistic Regression

- Loss function used is Categorical Cross Entropy Loss:

$$TrainLoss(\boldsymbol{w}) = -\frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{K}[y_{ik}log(p_{ik})]$$

- Gradient of the loss function:

$$\nabla_{\boldsymbol{W}} TrainLoss = \frac{1}{B}\boldsymbol{X}^{\boldsymbol{T}}(\boldsymbol{P} - \boldsymbol{Y})$$

- Update Rule:

$$\boldsymbol{W}_{new} = \boldsymbol{W}_{old} - \frac{\alpha}{B}\boldsymbol{X}^{\boldsymbol{T}}(\boldsymbol{P} - \boldsymbol{Y})$$

Classification Model Performance Evaluation

- Confusion Matrix: Table summarizing TP, FP, FN and TN.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

- Accuracy: Proportion of correct predictions.
- Precision: Proportion of true positives among predicted positives.
- Recall: Proportion of true positives among actual positives.
- F1 Score: Harmonic mean of precision and recall.

Formulas for Evaluation Metrics

$$\boldsymbol{Accuracy} = \frac{\boldsymbol{TP} + \boldsymbol{TN}}{\boldsymbol{TP} + \boldsymbol{TN} + \boldsymbol{FP} + \boldsymbol{FN}}$$

- Accuracy can be misleading with class imbalance

$$\boldsymbol{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FP}}$$

- Among predicted positives, how many are correct.

- Important when false positives are costly

Formulas for Evaluation Metrics

$$\boldsymbol{Recall = \frac{TP}{TP + FN}}$$

$$\boldsymbol{Recall \ = \ Sensitivity \ = \ TPR}$$

$$\boldsymbol{FNR = 1 - TPR}$$

- Among actual positives, how many are detected.
- Important when false negatives are costly

Formulas for Evaluation Metrics

$$\boldsymbol{F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall}}$$

- Ranges between 0 and 1.
- The closer to 1 the better.
- Balances precision and recall.

Formulas for Evaluation Metrics

$$\boldsymbol{Specificity = \frac{TN}{TN + FP}}$$

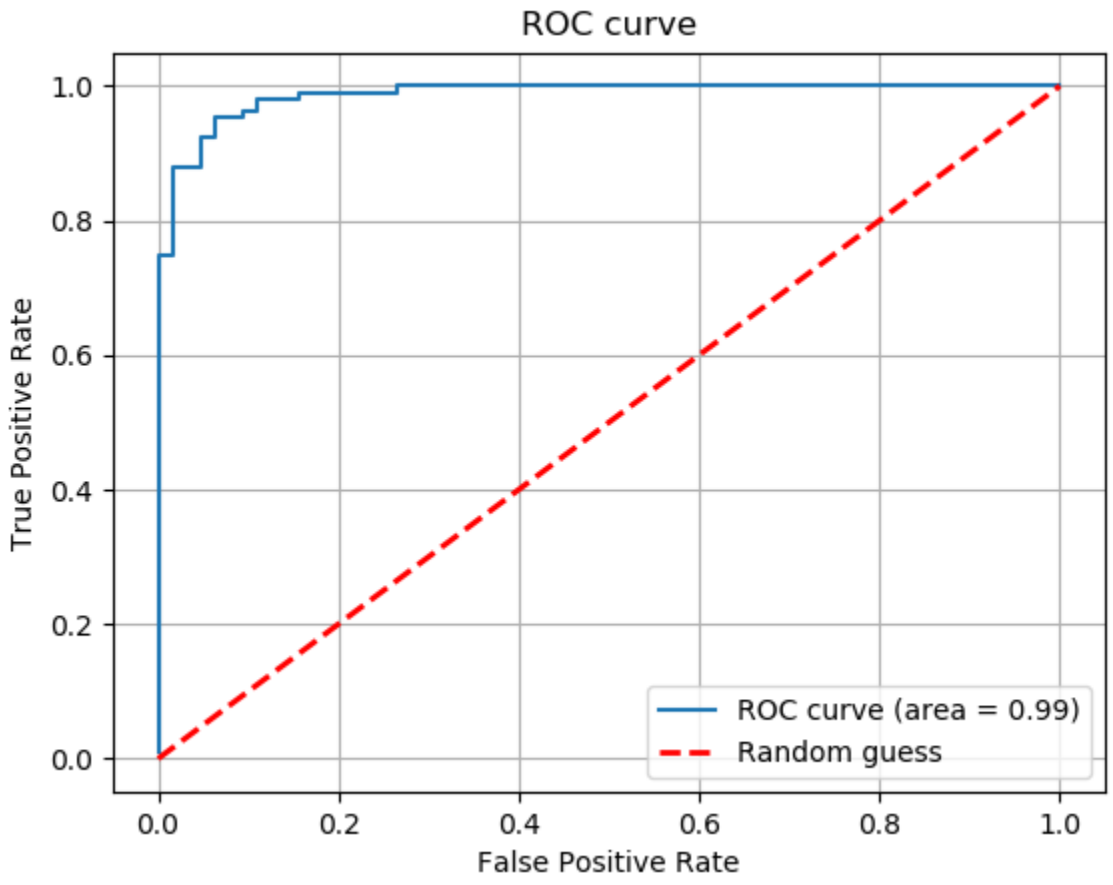
- Among actual negatives, how many are correctly rejected.

$$\boldsymbol{Specificity = TNR}$$

$$\boldsymbol{FPR = 1 - TNR}$$

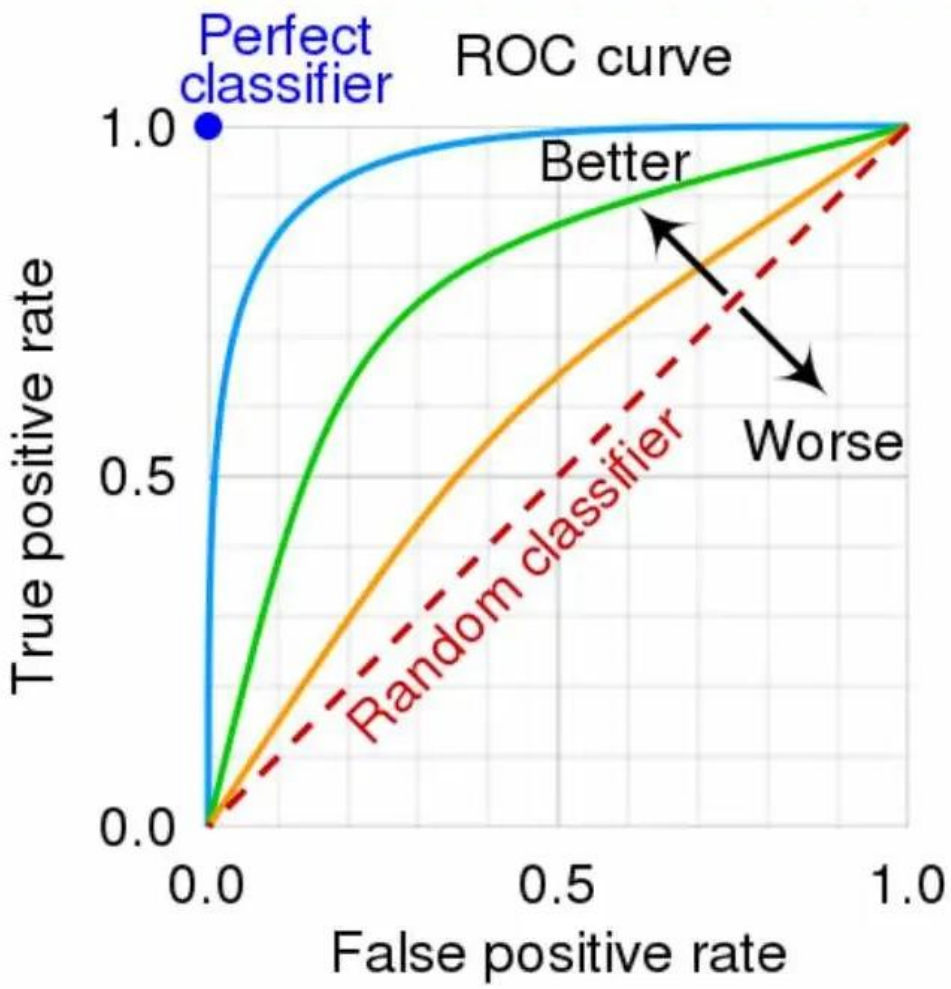
ROC Curve

- Plots TPR (Sensitivity) vs FPR (1-Specificity)
- Constructed by varying threshold at inference time

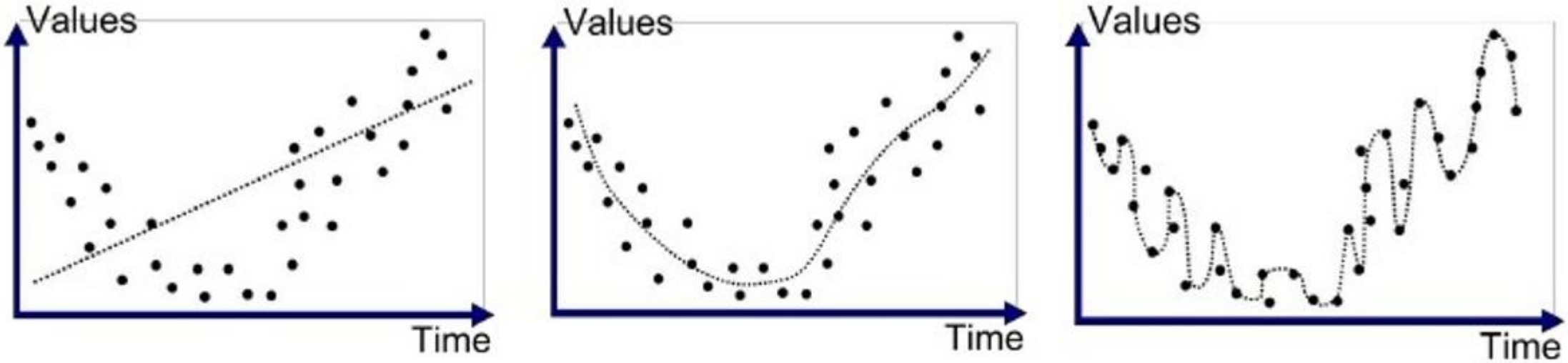


AUC (Area under the Curve)

- Summarizes ROC Curve Performance.
- The greater the AUC the better.



Underfitting and Overfitting



Underfitted

Good Fit/Robust

Overfitted

Underfitting and Overfitting

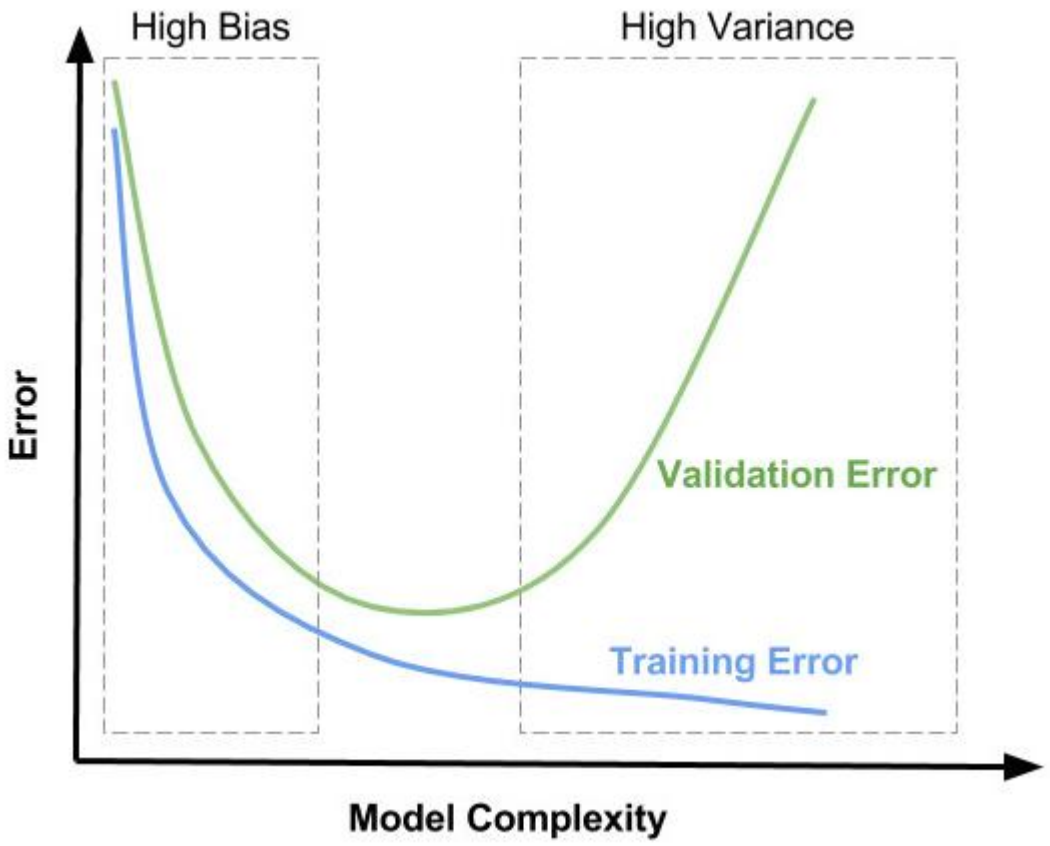
- Underfitted model is one in which a simple hypothesis class is taken for more complex data
- Underfitted models have high training and testing errors
- Overfitted model is one in which a complex hypothesis class is taken for relatively simpler data
- Overfitted models have low training errors but high testing errors

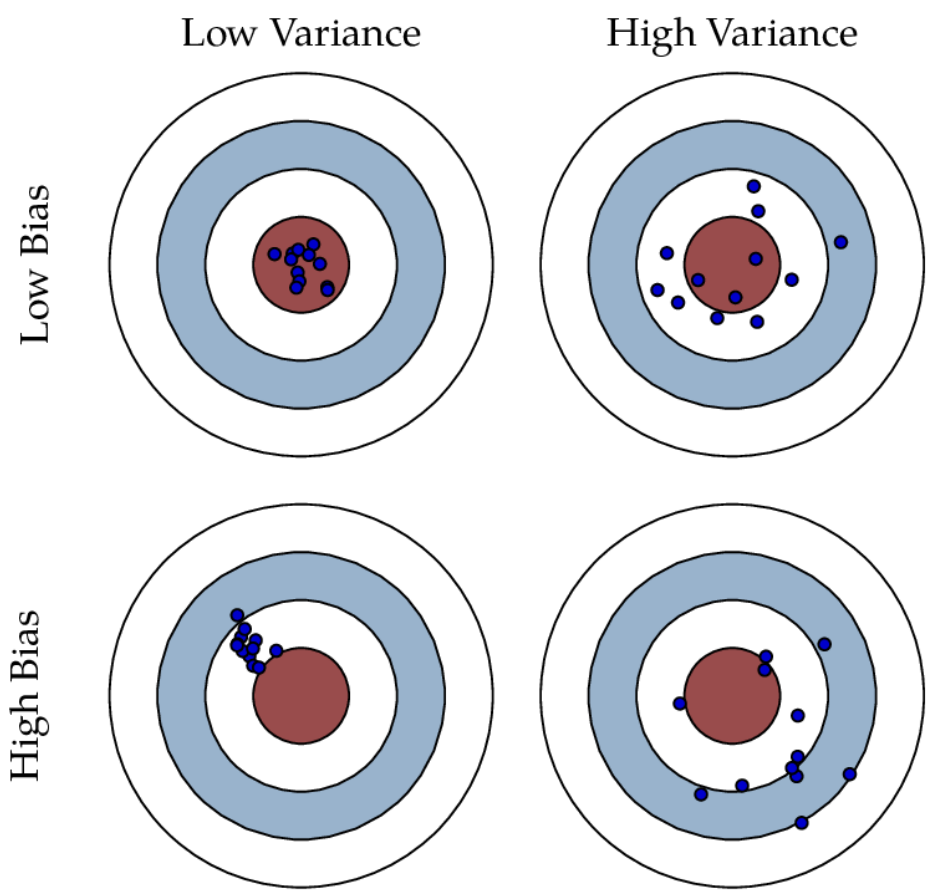
Bias and Variance

- Bias occurs when a model has limited flexibility to learn from data
- Overly simplified models result in high bias
- Training error and testing error follow similar trend
- Give high error on both training and test data
- High bias leads to underfitting

- Variance is an algorithm's sensitivity to specific sets of data
- Complex models result in high variance
- Pays a lot of attention on training data
- Small training error and large prediction error
- High variance leads to overfitting

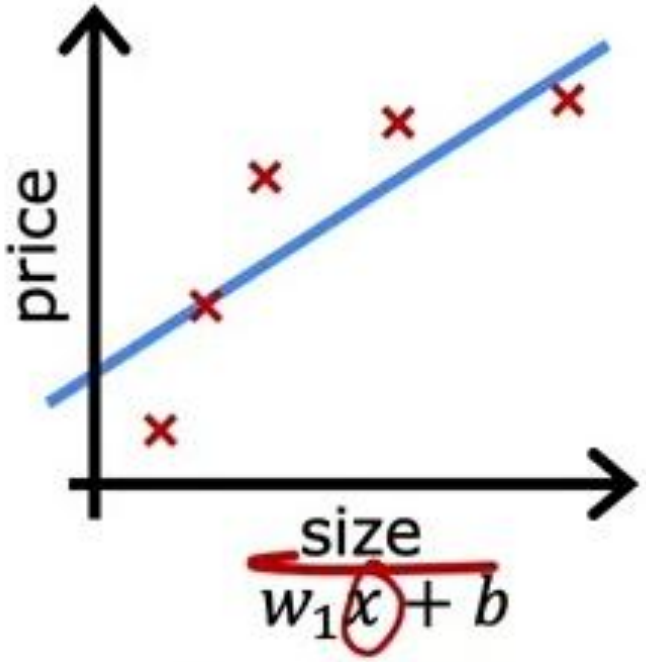
Bias and Variance





Regression Example

- Underfitted model

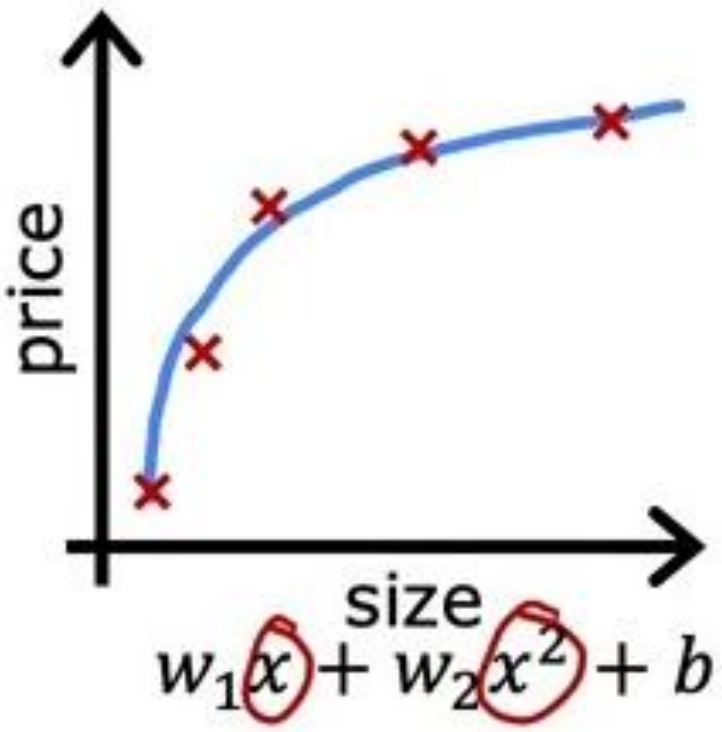


- Has high bias

- Learning algorithm has a strong preconception, or a strong bias, that the housing prices are going to be a completely linear function of the size despite data to the contrary.

Regression Example

- Fits training set well

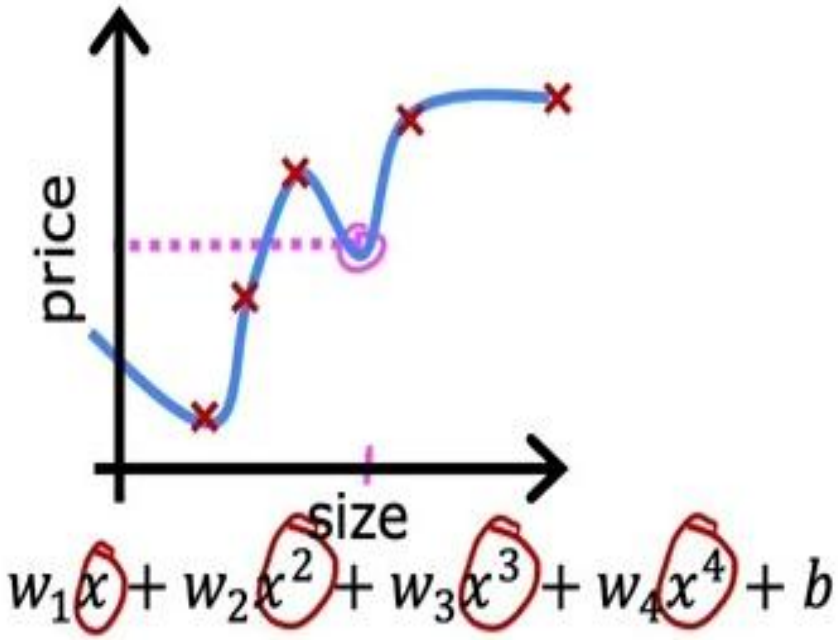


- Model generalizes well

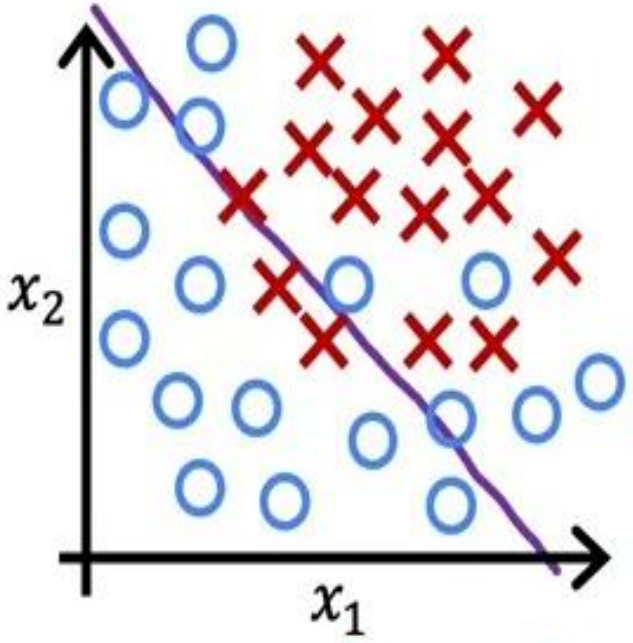
- Learning algorithm would make good predictions even on new examples that it has never seen before

Regression Example

- Overfitted Model
- Has high variance
- Fits the training set extremely well
- Learning algorithm results in zero loss on training set but would give high errors when predicting brand new examples



Classification Example

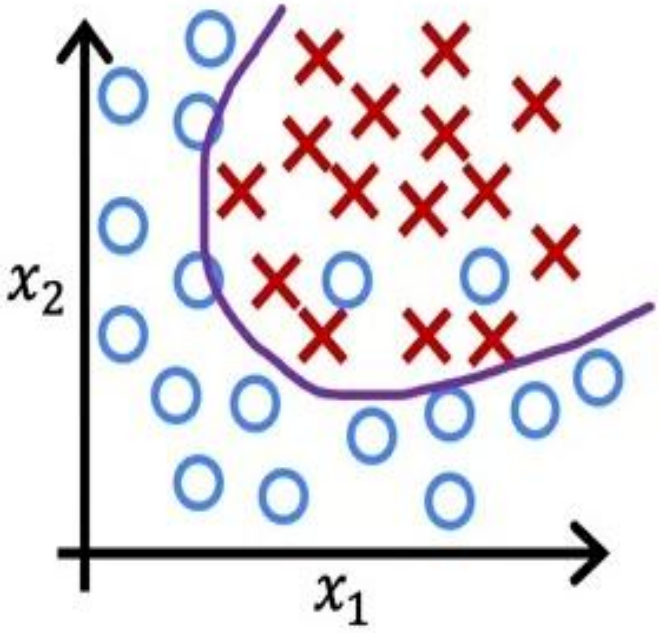


$$z = w_1 x_1 + w_2 x_2 + b$$

$$f_{\vec{w},b}(\vec{x}) = g(z)$$

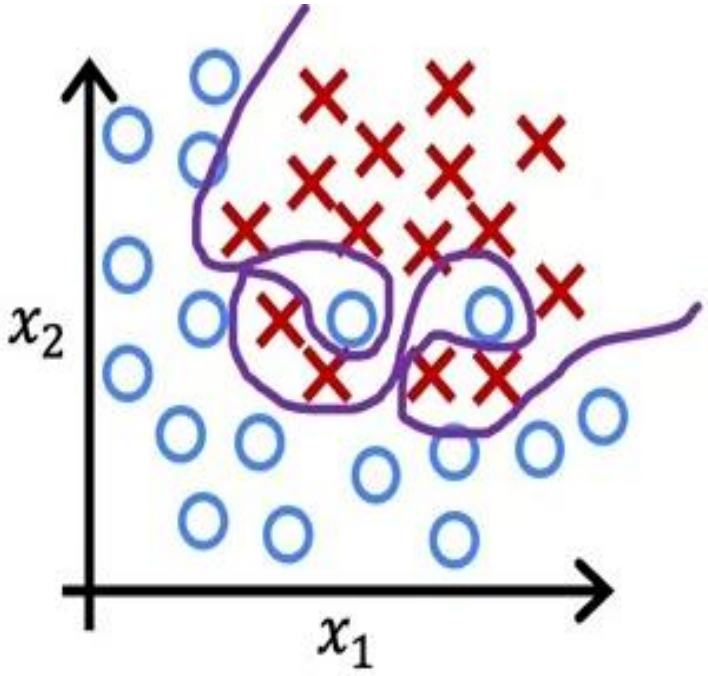
g is the sigmoid function

underfit high bias



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + b$$

just right



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + w_6 x_1^3 x_2 + \cdots + b$$

overfit

Questions?